



DEALER BEST PRACTICE SERIES

Airless Spray Painting

Application Maintenance Site Management Component Rebuild Safety MARC Management

Airless Spray Painting 0

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1.0 Introduction

Almost every repaired or rebuilt component must be painted. A durable coat of paint provides:

- An easy to clean and dirt repelling product surface.
- Protection from the typical Caterpillar operating environment
- The ability to help identify any unexpected leaks.
- A consistent product image.
- A professional finish that customers expect for a premium service.

In traditional dealer operations, components/machines are typically painted immediately after assembly/testing and cleaning.

In the more progressive service operations, piece parts are given an initial coat of paint early in the repair/rebuild process. Once medium/large size parts are reconditioned, exterior surfaces area painted while carefully shielding the interior surfaces. These parts, such as final drive spindles, wheels, brakes, and engine blocks, are much easier to keep clean through the repair/rebuild process.

An effective painting system provides:

- Professional painted surface finishes
- Minimal set-up and clean-up labor
- Multiple user flexibility
- Minimum supply/material (chemical) costs.

Airless spray systems provide the maximum flexibility at minimum cost of production.

2.0 Best Practice Description

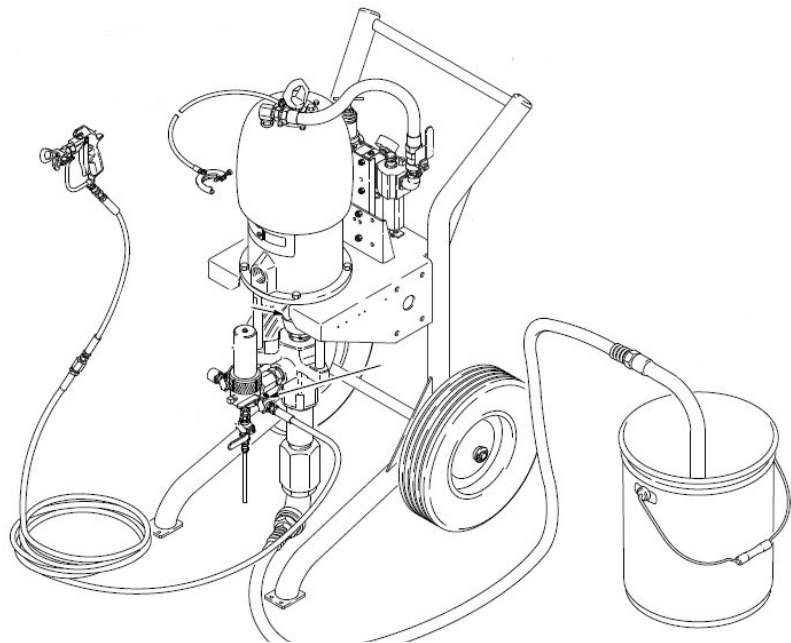
Painting operations are often not a core competency of a repair/rebuild shop. For that reason, it is often assigned to every technician and/or it is not provided effective tools/equipment or process.

Most shop managers would prefer to outsource this activity, but since painting is needed on an inconsistent basis, it is usually performed by the lower skilled, or all technicians.

Traditional air spray guns are typically used with a gun cup. More progressive shops use a remote paint tank with an air spray gun. Both arrangements require thinning the paint and filtering it before loading the cup or tank. The gun cup arrangement requires frequent cleaning (daily at minimum) and many shops deal with unclean spray guns that have hardened overnight and require extensive cleaning or replacement. A remote paint tank does not require as much cleaning making it well suited for the automotive industry which demands a higher grade finish, but may be over-kill (inefficient) for industrial products.

Airless painting provides an industry acceptable finish without the air spray process complexity. It is best suited for frequently used colors (typically Cat™ yellow). Other colors are best handled with aerosol cans or traditional air spray cups, given their limited application. In addition to being easier to manage, airless spray painting typically takes less time and covers surface

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Airless painting system showing a portable pump, bucket, and spray nozzle

imperfections better because each coat has more pigments (its thicker). Airless spray painting requires a slightly different technique. The painter must paint faster since the paint is flowing from the gun faster, which results in near show-quality finishes. Airless painting is especially effective in intermittent painting scenarios; typical in a rebuild center that only paints their rebuilt products.



Airless painting cart in a CRC environment

3.0 Implementation Steps

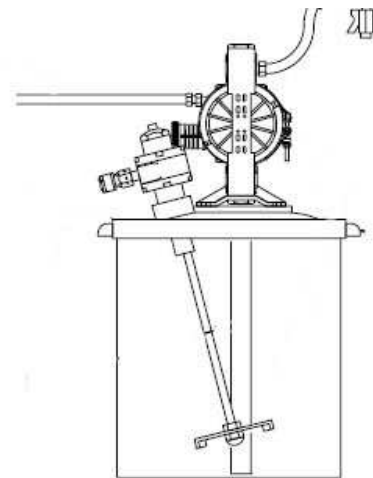
- Determine the anticipated type and volume of painting to be processed annually and the forecast for future volumes for at least the next five years.
- Establish painting equipment needs, placement, and storage requirements.
 - Research airless spray equipment suppliers and support providers.
 - Establish air spay equipment suppliers and support as needed.

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- Research Caterpillar equipment recommendations (airless equipment experience is with Graco).
- Determine support equipment/supplies needed for each spray technology. Include paint quantities, workbenches, storage cabinets, thinners, filters, replacement parts, and reference materials.
- Develop, adjust and document rebuild procedure with quality checklists as needed. Include “intermediate” parts painting if this is a preferred improvement.
- Establish new and adjusted rebuild time requirement targets as needed. These targets should decrease as well as supply consumption, such as thinning/cleaning chemicals. The savings should be taken in-house as a return on investment, not passed on to the customer.
- Present strategy to shop employees.
- Install the equipment changes into the shop (This may occur in phases as needed.)
- Train employees to use the new procedures/equipment. Follow-up/enforce immediately.
- Review process/equipment performance once established.
- Establish and install any adjustments as needed.

4.0 Benefits

- Increased capacity – Airless spraying reduces spray time with faster flowing paint sprays, and applying more paint per coat (fewer coats required). Paint dries faster because it is not thinned (the thinner doesn’t need time to evaporate out of the paint).
- Reduced labor – Several traditional painting steps are eliminated:
 - The equipment does not need daily cleaning. Many shops do not clean their airless equipment for months at a time. They typically shut the unit off and only make sure the spray tip is immersed in thinner to prevent dry-out.
 - The paint does not require thinning/straining. The equipment sprays pure paint at high pressure (>1000 psi). Inline electric heaters thin the paint if painting is performed in a cold environment <60° F. Debris is filtered out with an equipment standard inline filter. The filter is typically replaced annually or less.
 - The paint does not need mixing. A paint recirculation circuit can mix the paint. An optional lid (replacing the 5-gallon paint bucket) includes a mixing propeller and air-powered motor. The paint never needs handling, outside of opening the paint bucket and installing the suction-tube/mixer lid.
 - The airless spray unit can be installed outside of the paint booth, with only the spray lines/gun inside the booth. This facilitates easier booth cleaning and reduces equipment contamination/overspray/cleaning.
- Painting flexibility – The paint equipment needs only to be switched on to begin painting. If the paint has been settling for more than a day, the operator may need to turn the mixer on for



A covered paint bucket with mixer

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5 minutes prior to painting (while they prepare their piece for painting). Once painted, the operator, switches the equipment off, and immerses the spray tip into thinner. There is no need to batch-process components. An airless painter is portable and can be transported for field painting where the procedures are greatly simplified for a dirtier environment.

- Improved operator safety – Although paint is sprayed at >100psi (can cause injury), this is easily avoided with reasonable caution. Paint operators receive much less chemical exposure since they do much less paint/thinner handling and less equipment cleaning.
- Reduced cost – Less paint thinners, filters, and cleaners are used, reducing supply costs. Airless painting produces less cleaning rags and thinners for disposal.

5.0 Resources Required

- Investment Costs Vary (\$3,000 - \$4,000 US) – airless spray systems could include:
 - High-pressure pumps (such as the Graco Bulldog or King models)*
 - Inline paint filter*
 - Paint container lid with propeller mixer*
 - Inline electric paint heater (typically not needed)
 - * Recommended for initial painting unit.
- Additional equipment for non-yellow paint (dealer often has this already):
 - Air spray gun cups or remote paint cups.
 - Paint mixing/straining equipment
- Support Equipment
 - Workbenches and supply shelves.
 - Dry waste and fluid waste containers.
- People
 - Establish current painting/cost concerns with team.
 - Establish process, layout, and equipment improvements through team.
- Training
 - Establish the improvement benefits with the production team.
 - Define, establish, and enforce the new process and duties with the production team.

6.0 Supporting Attachments / References

Diagrams shown from:

Graco Inc.
88-11th Avenue N.E.
Minneapolis, MN 55413
www.graco.com

7.0 Related Best Practices

None.

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8.0 Acknowledgements

This Best Practice was written by:

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